

SynthProof — synthetic data that ships with a checkable proof

Differentially private synthetic data that keeps the real patterns, leaks no membership, and — uniquely — carries a **signed release a stranger can verify from the artifact alone**.

THE PROBLEM (MEASURED, NOT HYPOTHETICAL)

0/290

HuggingFace synthetic datasets declare DP with an ϵ

0/12

flagship real DP deployments are signed or machine-checkable

Practitioners don't verify DP — they trust blindly (Song, CSCW'24)

WHAT WE BUILT

1 · Boundary auditor

RB1–RB14: reads only the artifact, flags every channel that leaks outside ϵ (seed, row count, fingerprint, domain, multi-table...).

2 · Signed checkable label

Dibia's 9 fields + Ed25519 signature + operating-range. Conformance = passing the auditor. MLCommons-validated.

3 · Gated pipeline

synthesise → sign → Croissant → assert all gates → one report. The runnable artifact an examiner can run.

RESULTS ON THE FULL DATASETS (EVERY ROW)

Adult

30,162

0.0072

corr err

92%

of real-data ML

ACS

195,665

0.0017

corr err

89%

of real-data ML

Bank

45,211

0.0009

corr err

90%

of real-data ML

Diabetes

99,492

0.1185

corr err

99%

of real-data ML

Privacy audit — membership attacker $\approx$ coin-flip everywhere (no re-identification). Canary audited $\epsilon = 0$ is the tool's *floor*; our GDP audit recovers a real signal: **independent0.36**
aim**0.18** mst**0.15** (reported honestly as a floor, not "zero leakage").

HOW WE'RE BETTER THAN THE LITERATURE

vs Dibia (P10)

They *proposed* a DP label. **We built it** + signed it + report its limits.

vs Ganev / Census (P4,P5,P9)

Their leaks live in papers/PDFs. **We make them a machine-checkable field** (RB6/RB9/RB8).

vs Annamalai (P2)

Audited ϵ reads 0. **We report it as a floor, not 'safe'** — and recover a real GDP signal.

vs McKenna AIM (P8)

We don't beat AIM — **we wrap it in a signed, checkable release** it never had.

vs Song (P11)

They found nobody verifies DP. **We ship the checker** a stranger can run.

THE ONE AXIS WHERE WE CLEARLY LEAD

14

release-boundary checks (RB1–RB14) — the tool none of them ship

1st

signed + machine-checkable DP synthetic-data release label

0 fab.

fabricated numbers — every figure reproducible, limits reported

Honest bottom line: we don't claim a better generator than AIM or the Ganev audits — we build on them. Our contribution is the **signed, machine-checkable release label + checker** that Dibia et al. called for and nobody ships, shown against **4 real datasets** (census · finance · healthcare) where the synthetic data reaches up to **92% of real-data ML utility** with **no detectable membership leak**.